



Office of the Science and Technology Adviser to the Secretary

U.S. DEPARTMENT of STATE

International Semiconductor R&D Capabilities Mapping

Report and Database Presentation

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Introduction



Goals and Scoping

Goals

- **Map global semiconductor R&D**, highlighting technology focus areas and institutional research capabilities.
- **Develop a methodology** for research, analysis, and presentation for future continuation and adaptation.

Scope

- Focus on **less analyzed countries** conducting high-impact research to highlight the global ecosystem.
- Largely avoid work conducted in **China** or the **United States** to avoid duplicating existing projects and reasonably scope research.



Classification Criteria

Value Chain Focus

- **Design** (materials design, architectural design, and security and IP)
- **Manufacturing and Assembly** (advanced packaging and advanced design)
- **Applications** (AI and computing, energy, manufacturing, and defense)
- **Basic Research** (advanced materials and heterogeneous applications).

Research Capabilities

- Modeling and simulation and design
- Fabrication
- Wafer and die measurement and metrology
- End-use integration and test
- Other capabilities

Research Areas

- Applications
- Emerging technologies
- Design
- Architecture
- Storage and memory
- Fabrication and manufacturing
- AI and machine learning

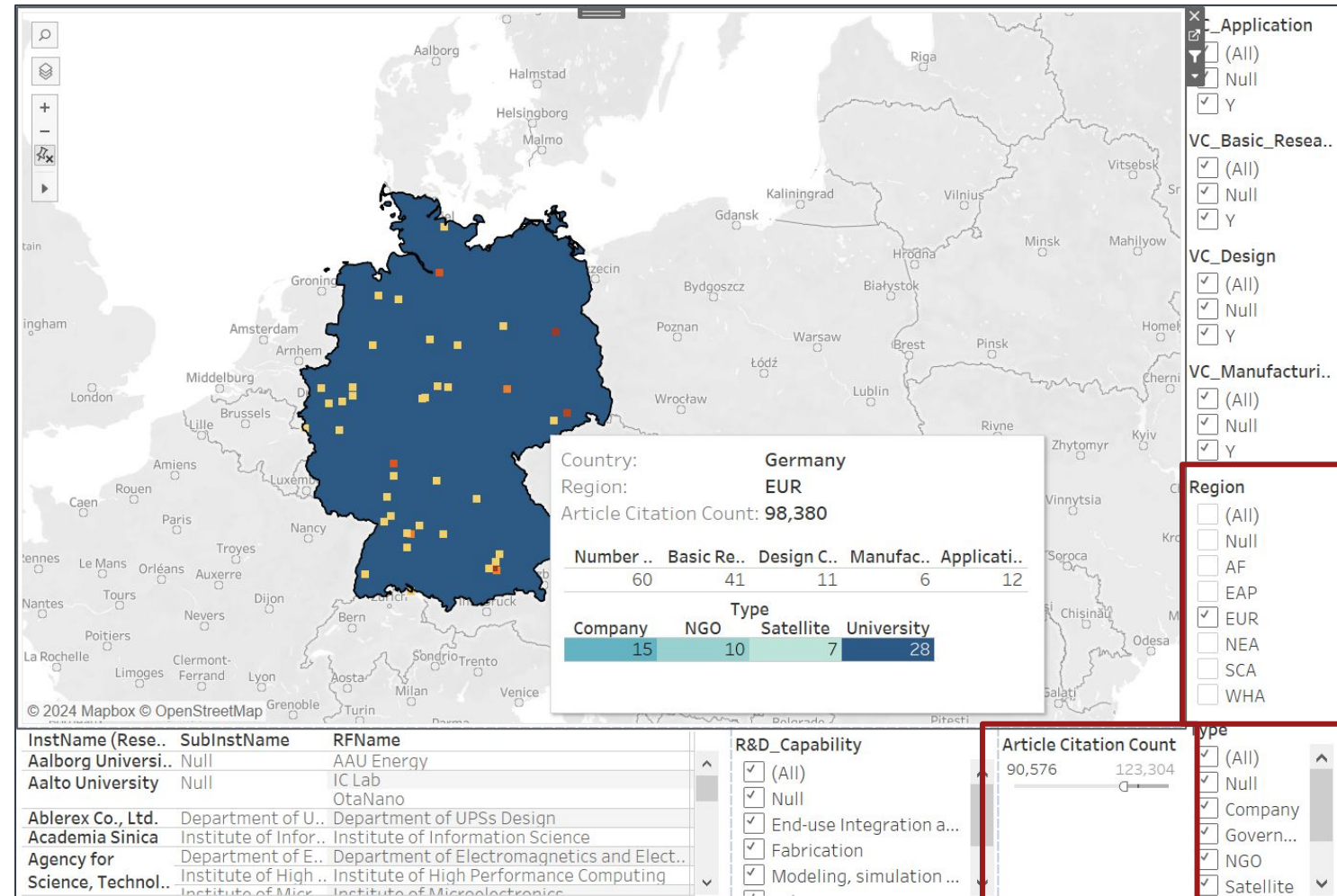


Database and Dashboard



Dashboard Preview: Country Level

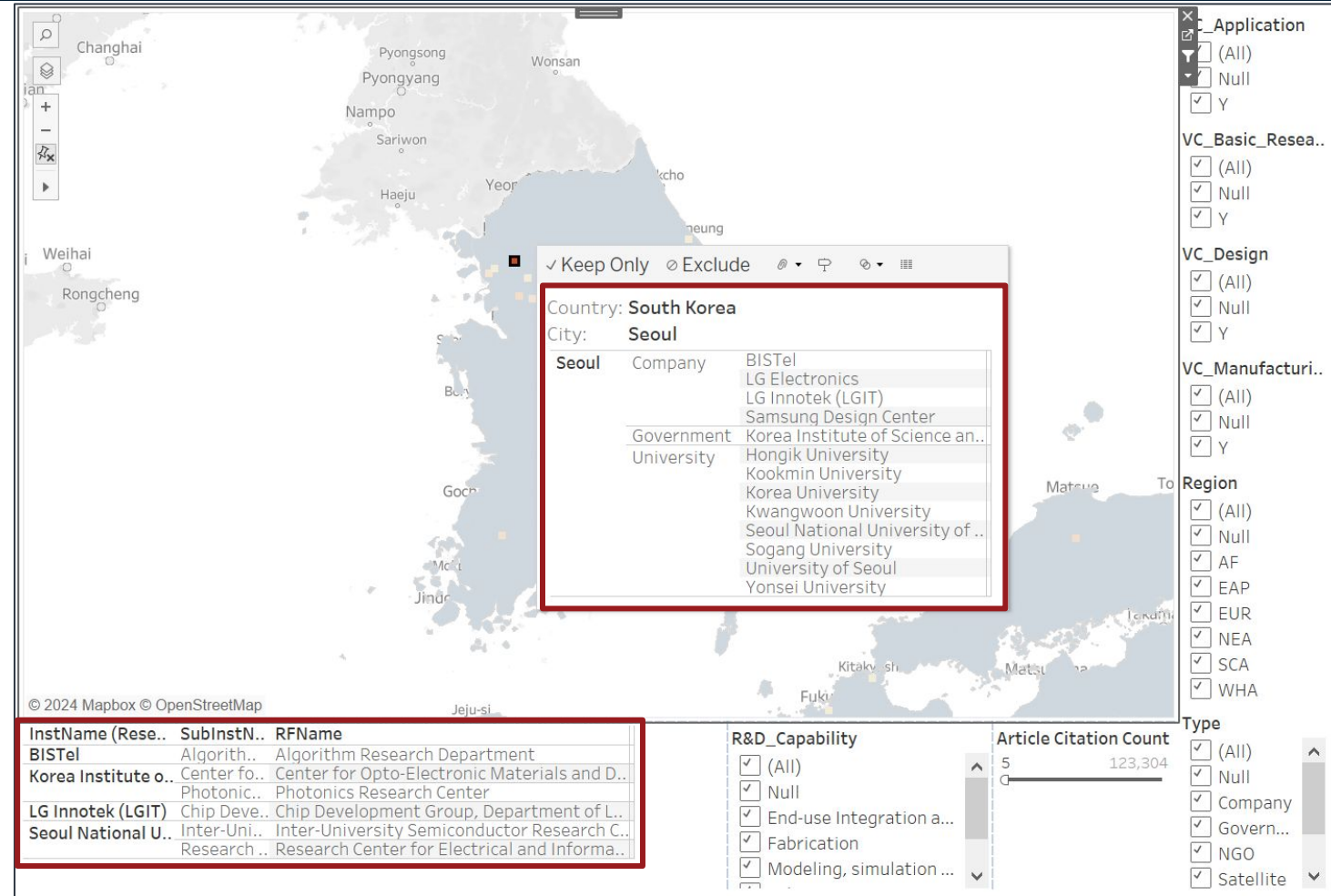
- What if we're interested in top-performing countries in EUR?
 - Easily filter by region and aggregate citation count
 - We find that Germany is the top country in EUR with 60 institutions





Dashboard Preview: City and Institution Level

- What if we want to know more about the ecosystem in Seoul, SK?
 - Can get a breakdown of institutions by type
 - Also see a filtered view of relevant subinstitutions and research facilities
 - Live demo later!





Where did the data come from?

- Institute of Electrical and Electronics Engineers (IEEE) Xplore
 - Top global publisher of electrical engineering research
 - approx. $\frac{1}{3}$ globally
 - Global in scope
 - Resources for authors from non-English speaking/smaller R&D ecosystem countries
 - Relatively simple to filter and export results
 - Some automation required to speed up process
- United States Patent Office (USPTO)
 - Global patents require US patents to be covered in the US
 - Patent Cooperation Treaty (PCT) provides reciprocal coverage in the US for certain international patents
 - Free API to retrieve data
- Manual Research
 - Automatic data retrieval wasn't enough to get as granular as we wanted
 - Used IEEE/USPTO data as a foundation for manual research
 - Used custom Excel template to enforce data standardization



IEEE

IEEE Xplore®

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Can easily filter by year,
keywords and type of
publication

Can export search results but is
limited - automated web
browsing tools were used to
download multiple tranches at
once

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Showing 1-100 of 46,091 results for **semiconductor** ×

▾ Filters Applied: **Journals** × **2000 - 2024** ×

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☐ Magazines (2,114)

☐ Books (822)

☐ Early Access Articles (453)

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PURPOSE

LEGACY API FAQs

LEGACY API QUERY LANGUAGE

LEGACY API ENDPOINTS

REQUEST AN API KEY

PATENTS (LEGACY)

INVENTORS (LEGACY)

ASSIGNEES (LEGACY)

LOCATIONS (LEGACY)

CPC (LEGACY)

USPC (LEGACY)

NBER (LEGACY)

Patents Endpoint

Patent Field List

GET /api/patents/query?q{f,o,s}

POST /api/patents/query

- Used legacy API (new API was still in development)
- Can find query structure, logical operators and list of fields available for query on USPTO website

Patent Field List

Filter by field name..

API Field Name	Group	Common Name	Type	Query	Description
appcit_app_number	application_citations	Application Number	string	Y	Application ID (issued by USPTO) for application cited by the selected patent

- Our query:
 - Checked abstract and title for semiconductor
 - Also used with more specific keywords
 - Returned basic identifying information for patent and inventing organization
 - Also included various other data points we considered using as measures of patent value

```
query = {'q': {'_and': [  
    # {'_or': [  
        # [{'_text_any': {'patent_abstract': f'{abstract_search_term}'},  
        #   {'_text_any': {'patent_title': f'{abstract_search_term}'}}  
        ],  
        # {'_or': [  
            # [{'_text_any': {'patent_abstract': 'semiconductor'},  
            #   {'_text_any': {'patent_title': 'semiconductor'}}  
            ],  
            },  
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        ]},  
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```



Data Types

- Three tiers of information collected for each institution
 - Tier I
 - Primary identifying information
 - Tier II
 - Characteristic information of the institution and research
 - Tier III
 - Granular details like specific equipment, funding sources and collaborations
- Tier I/II involved mix of automatic and manual collection

Tier	Data Point	Source
Tier I	Institution Name	Provided from IEEE/USPTO data
	State/Province	Manual research
	City	Manual research
Tier II	Sub-institutions (colleges, etc.) (list)	Provided from IEEE/USPTO data or from manual research
	Research facilities (list)	Provided from IEEE/USPTO data or from manual research
	Research areas (list)	Manual research and classification
	Value chain	Manual research
	Institution type	Manual research and classification
	Contact information	Manual research
Tier III	Funding sources, collaboration types, amounts, and collaborations (list)	Manual research and classification
	R&D capabilities (e.g., design, fabrication, modeling, simulation, etc.)	Manual research and classification



Manual Data Collection

- Automatic data collection was mostly done through web scraping
- Manual data collection relied on information gathering from institutional websites, as well as decision-making about where to categorize institutions once information was gathered
- Throughout, we adhered to a strict classification schema that was compatible with the creation of the map

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	R&D capabilities (e.g., design, fabrication, modeling, simulation, etc.)	Manual research and classification



Live Dashboard Demo



Report and One-Pagers



Global Snapshot



- 624 institutions, including 360 universities, 180 companies, 54 non-profits, and 30 government
- Institutions held diverse roles across the value chain, including 427 in basic research, 244 in applications, 211 in design, and 101 in manufacturing



Global Snapshot

Figure 1: Global breakdown of institutions by research area

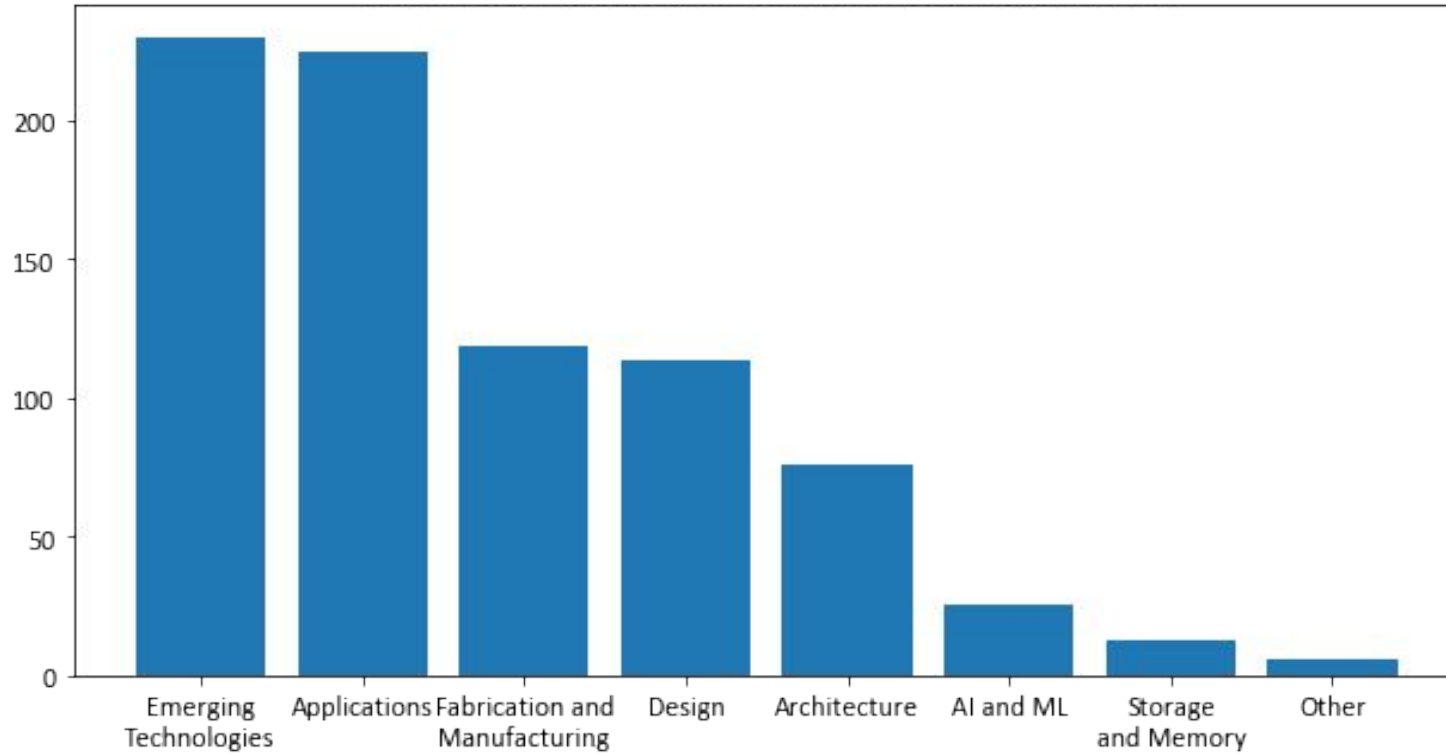
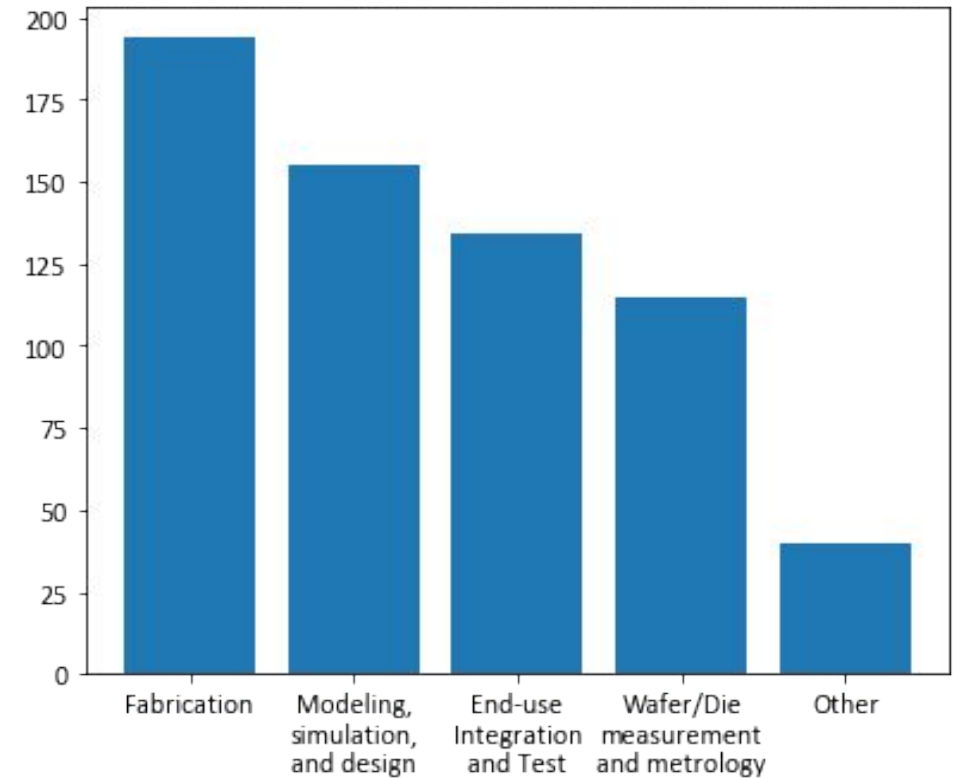


Figure 2: Global breakdown of institutions by equipment type





Regional Trends

- EUR and EAP account for the majority (84.6%) of total institutions
- Regions tend to specialize across the value chain
- EUR leads in the amount of research areas

	AF	EAP	EUR	NEA	SCA	WHA
End-use integration and testing	2	16	95	2	7	13
Fabrication	1	28	145	1	9	15
Modeling, simulation, and design	0	60	54	5	17	18
Wafer/die measurement and metrology	0	1	93	1	10	11



Clusters

- **Clusters of innovation** - areas exhibiting high interconnectedness between universities, industry, and other stakeholders

Country	City	# of Institutions	Country	City	# of Institutions
Japan	Tokyo	27	India	Bangalore	5
France	Paris	19	Germany	Berlin	5
South Korea	Seoul	15	Ireland	Dublin	5
Singapore	Singapore	14	South Korea	Icheon	5
Taiwan	Hsinchu	12	Japan	Kanagawa	5
France	Grenoble	7	Taiwan	Kaohsiung	5
United Kingdom	London	7	Japan	Kawasaki	5
Canada	Ottawa	7	Australia	Sydney	5
South Korea	Daejeon	6	France	Toulouse	5
South Korea	Suwon	6			